

# Generation Interconnection Request System Impact Study Report

## For Queue Project Q22-2025

Agua Fria 138 kV  
January 22, 2024

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2025. The Interconnection Customer proposes a 108.0 MW solar photovoltaic facility interconnecting at the Agua Fria 138 kV substation in the Pennsylvania area. The request was received on March 06, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 499 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 14 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$101,322,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

| Monitored Facility                                      | Contingency                             | Loading (%) | Rating (MVA) | DFAX (%) |
|---------------------------------------------------------|-----------------------------------------|-------------|--------------|----------|
| Cypress Gardens 161/138kV Transformer Bank              | Wimbledon Substation-Blount St 138 kV   | 171.2       | 96           | 6.3      |
| San Simeon-Stefano Substation-Puntledge 138 kV          | Jh Warden-West Temple 345 kV            | 125.8       | 195          | 33.5     |
| Arecibo Pueblo 8002-Wimbledon 138 kV L                  | Chambers Springs-Jh Warden 500 kV       | 121.4       | 201          | 6.7      |
| Cisco Grve 161-138 kV Autotransformer                   | Strawberry Belt-Jh Warden 345 kV        | 119.5       | 106          | 18.9     |
| New Plymouth 161/138 kV Transformer                     | Agua Fria-New Plymouth 138 kV L         | 116.6       | 115          | 43.9     |
| Puntledge-New Plymouth 138 kV                           | West Temple-Jh Waren 345 kV line        | 115.5       | 222          | 33.5     |
| Blount Street-Wimbledon 138 kV L                        | Jh Warden-Chambers Springs 500 kV L     | 114.3       | 226          | 6.7      |
| Lawton East Side-West Temple 138 kV Line                | Lancaster-Johnson 138 kV                | 110.7       | 249          | 5.5      |
| Blount St-Cypress Gardens 138 kV                        | Jh Warden-Chambers Sprngs 500 kV        | 109.5       | 233          | 6.7      |
| San Simeon-Stefano 161-138 kV Autotransformer           | Highway Jct 1-San Simeon-Stefano 138 kV | 105.8       | 108          | 26.9     |
| Highway Junction 1 Substation-San Simeon-Stefano 138 kV | West Temple 345/138 kV Transformer Bank | 102.0       | 291          | 22.6     |
| Hills Chapel-Wallenpaupack 345 kV                       | W Gallatin-Walnut 345 kV                | 101.9       | 1,817        | 5.0      |
| Coronado Substation-Burns 345 kV                        | Clear Springs 500 kV/345 kV XFMR        | 101.1       | 1,655        | 5.3      |
| Ontelaunee Energy Ctr-Wigwam Substation 345 kV Line     | Fayette-Ironwood Llc 345 kV             | 100.0       | 1,307        | 5.8      |

### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

## 4. Required Network Upgrades and Cost Estimates

| Facility                                            | Required Upgrade                            | Cost Estimate | Allocation to Project |
|-----------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| Cypress Gardens 161/138 kV Autotransformer          | Replace transformer with higher-rated unit  | \$6,247,000   | 100% (\$6,247,000)    |
| San Simeon-Stefano Substation-Puntledge 138 kV      | Rebuild line with higher-capacity conductor | \$5,824,000   | 100% (\$5,824,000)    |
| Arecibo Pueblo 8002-Wimbledon Substation 138 kV     | Rebuild line with higher-capacity conductor | \$12,501,000  | 100% (\$12,501,000)   |
| Cisco Grove 161 kV/138 kV Transformer               | Replace transformer with higher-rated unit  | \$4,619,000   | 33% (\$1,544,000)     |
| New Plymouth 161-138 kV Transformer Bank            | Replace transformer with higher-rated unit  | \$8,101,000   | 100% (\$8,101,000)    |
| Puntledge-New Plymouth 138 kV                       | Rebuild line with higher-capacity conductor | \$42,167,000  | 100% (\$42,167,000)   |
| Blount Street-Wimbledon 138 kV L                    | Rebuild line with higher-capacity conductor | \$12,157,000  | 100% (\$12,157,000)   |
| Lawton East Side Substation-W Temple 138 kV         | Rebuild line with higher-capacity conductor | \$2,738,000   | 41% (\$1,124,000)     |
| Cypress Gardens-Blount St 138 kV                    | Reconductor line                            | \$1,720,000   | 100% (\$1,720,000)    |
| San Simeon-Stefano Substation 161/138kV XFMR        | Replace transformer with higher-rated unit  | \$4,095,000   | 100% (\$4,095,000)    |
| Highway Junction 1-San Simeon-Stefano 138 kV line   | Reconductor line                            | \$1,283,000   | 66% (\$841,000)       |
| Hills Chapel-Wallenpaupack 345 kV                   | Reconductor line                            | \$11,378,000  | 23% (\$2,636,000)     |
| Burns-Coronado 345 kV L                             | Reconductor line                            | \$6,511,000   | 24% (\$1,562,000)     |
| Ontelaunee Energy Ctr-Wigwam Substation 345 kV Line | Reconductor line                            | \$11,397,000  | 7% (\$803,000)        |
|                                                     | Total allocated to project                  |               | \$101,322,000         |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.